

# The Role of Gravitational Perturbations in Determining Jupiter's Mass: A Comparative Study of Io and Callisto

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Kepler's third law provides a fundamental approach to estimating a planet's mass. It assumed an idealised system where perturbations are neglected, and celestial bodies are treated as point masses. This paper discusses how perturbations influence the accuracy of Jupiter's mass estimation determined using Kepler's third law by tracking the orbital motions of Io, its closest Galilean moon, and Callisto, its furthest. It was hypothesised that Callisto would yield a more accurate estimate, as it experiences weaker perturbative effects. Multiple telescope images of both moons were analysed using Gaia, and their distances from Jupiter were modelled to determine their semi-major axes and orbital periods. Using Kepler's third law, Jupiter's mass was estimated as  $(2.0 \pm 0.5) \times 10^{27}$  kg from Io and  $(1.91 \pm 0.03) \times 10^{27}$  kg from Callisto, compared to the accepted value of  $1.8981(8) \times 10^{27}$  kg. The analysis of perturbations, including solar influence, mutual satellite interactions, and Jupiter's oblateness, showed that Io is more affected by Jupiter's oblateness than Callisto. This supports the hypothesis that Callisto provides a more accurate mass estimate as it is less susceptible to perturbative influences.

## I. INTRODUCTION

Jupiter is the most massive planet in our solar system and has the second-highest number of natural satellites [10]. Among these, its four largest and most massive moons — Io, Europa, Ganymede, and Callisto — are collectively known as the Galilean moons [9].

Determining Jupiter's mass is crucial for space exploration, as precise calculations are essential for planning space probe trajectories, particularly during gravity assist manoeuvres to prevent unintended collisions. For instance, NASA's Voyager 1 probe, launched to study the outer planets, used Jupiter's gravity to gain speed on its journey toward Saturn, Uranus, and Neptune. However, accurately measuring Jupiter's mass using the orbits of the Galilean moons is challenging due to various perturbative effects.

The primary method to determine the mass of a planet with moons is to use Kepler's third law [13], which is expressed as

$$T^2 = \frac{4\pi^2 a^3}{G(M_1 + M_2)} \quad (1)$$

where  $T$ ,  $a$  and  $M_1$  are the orbital periods, semi-major axes and masses of its moons respectively and  $M_2$  is the mass of the planet. This law is derived from Newton's law of universal gravitation and the centripetal force acting on a body orbiting a spherically symmetric planet. It assumes an idealised two-body system in which celestial bodies are treated as point masses [12]. However, in reality, deviations arise due to planetary oblateness caused by rotation, solar perturbations, mutual interactions between moons, and gravitational influences from other celestial bodies. These factors can affect the mass estimation of a planet derived from Kepler's laws.

Since the masses of Io and Callisto which are used in this paper are negligibly small compared to that of Jupiter, the  $M_1$  term can be ignored, and equation (1) is simplified to

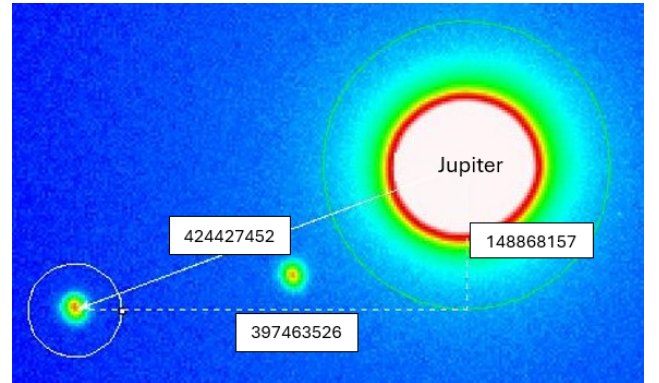
$$T^2 \approx \frac{4\pi^2 a^3}{GM_J} \quad (2)$$

where  $M_J$  is the mass of Jupiter. Rearranging equation (2) yields

$$M_J \approx \frac{4\pi^2 a^3}{GT^2} \quad (3)$$

This paper describes an investigation which hypothesises that Callisto, the most distant of Jupiter's Galilean moons, will yield a more precise estimate of Jupiter's mass than Io, its closest Galilean moon. Since Callisto orbits farther from Jupiter, it is less affected by Jupiter's oblateness and potential perturbations from orbital resonance. The perturbation due to the orbital resonance was initially considered to affect only Callisto, as it is not part of the Laplace resonance (orbital resonance with three celestial bodies), which involves Io, Europa, and Ganymede.

## II. METHODOLOGY



**FIG. 1:** The position of Io was determined using the StarLink website [6]. The distances are shown in metres.

A telescope with a pixel scale,  $p_s$  of 0.983 arcsecond per pixel [1] was used to capture multiple images of Jupiter and its Galilean moons over a two-week period to track the positions of Io and Callisto. To accurately identify Io and Callisto among the Galilean moons, an external website [6] was used, where the observation dates were input to determine their positions at the corresponding times. The images were processed using Gaia. The linear distance from Jupiter to each moon was measured in pixels by constructing a right triangle, where the hypotenuse extends from the centre of Jupiter to the respective moon, as shown in FIG. 1.

The measured distances (in pixels),  $d$  were converted to angular separations,  $\theta$  using the equation  $\theta = dp_s/206265$ . The separations,  $L$  between Jupiter and the moons were

then calculated using the equation

$$L = r\theta \quad (4)$$

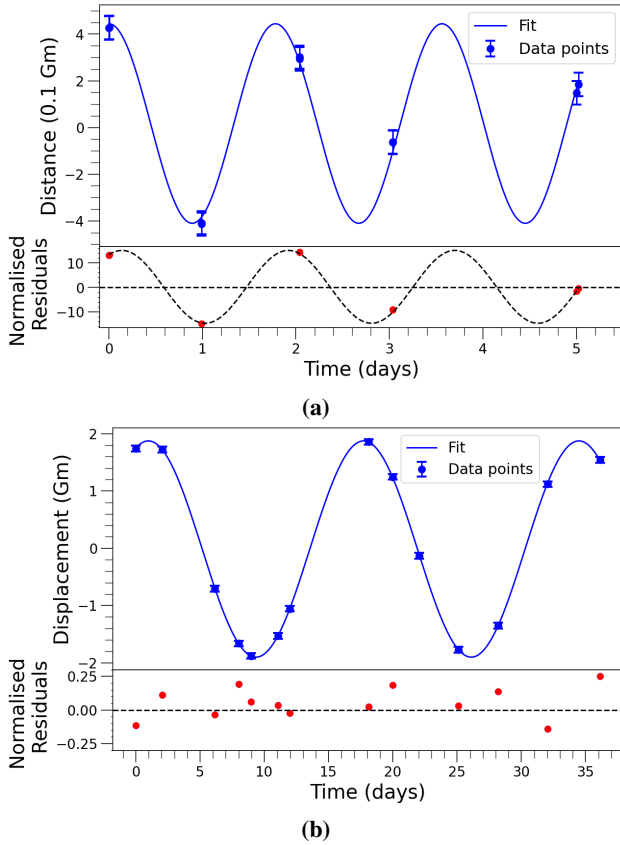
where  $r$  is the distance from Earth to Jupiter, which is obtained from [6] and small angle approximation was applied.

Graphs of the separation distances between Jupiter and its moons over time (in days) were plotted. The displacement of Io and Callisto is assumed to follow a periodic, approximately sinusoidal pattern as viewed from Earth, leading to the model:

$$f(t) = A \sin(Bt + C) + D \quad (5)$$

The semi-major axes,  $A$  and orbital periods,  $2\pi/B$  were obtained from the best-fit parameters and substituted into equation (3) to estimate the mass of Jupiter.

### III. RESULTS



**FIG. 2:** (a) Orbital motion graph of Io plotted using non-linear least square fit, with potential pattern in the residual plot. (b) Displacement-time graph of Callisto plotted using non-linear least square fit.

From FIG. 2(a), the orbital period of Io is determined to be  $(1.78 \pm 0.03)$  days, while for Callisto, FIG. 2(b) gives an orbital period of  $(16.76 \pm 0.04)$  days. The semi-major axes, obtained from the amplitudes of the graphs, are  $(4.3 \pm 0.3) \times 10^8$  m for Io and  $(1.89 \pm 0.01) \times 10^9$  m for Callisto. These values are in agreement with [9].

Substituting these values into equation (3), along with the gravitational constant  $G = 6.67 \times 10^{-11} \text{ m}^3 / \text{kg s}^2$ , yields Jupiter's mass as  $M_J^{(I)} = (2.0 \pm 0.5) \times 10^{27} \text{ kg}$  when using

Io's data and  $M_J^{(C)} = (1.91 \pm 0.03) \times 10^{27} \text{ kg}$  when using Callisto's data.

From FIG. 2(a), the reduced chi-squared value is  $\chi_v^2 = 0.46$  with a p-value of  $P(\chi_{min}^2, v) = 0.84$  where  $v$  is degree of freedom. Compared to an ideal fit, where  $\chi_v^2 \approx 1$  and  $P(\chi_{min}^2, v) \approx 0.5$ , this suggests that errors might be overestimated or that the model could be slightly overfitting. Additionally, a potential periodic pattern appears in the residuals, which may indicate perturbations or an insufficient number of data points. The later sections will quantify and determine whether this pattern arises from data insufficiency or periodic perturbations.

From FIG. 2(b), the reduced chi-squared value is  $\chi_v^2 = 0.082$  and a p-value of  $P(\chi_{min}^2, v) = 0.99$ , which strongly suggests overfitting due to an excess of free parameters,  $C$  and  $D$  which lack physical significance. While multiple models were tested, the current model offers the best balance between complexity and accuracy, yielding a better  $\chi_{min}^2$  and  $P(\chi_{min}^2, v)$ . This highlights the need for further investigation into error estimates and model refinement.

Aside from measurement uncertainties, the interactions between Jupiter and its moons introduce additional sources of error. The following section will examine these non-Keplerian influences to assess their impact on the accuracy of our mass determination.

### IV. ANALYSIS AND MODELLING PERTURBATIONS

#### A. Solar Perturbation

Solar perturbation was initially thought to have a significant impact on the calculation of Jupiter's mass due to the Sun's massive gravitational influence. To quantify this, the Hamiltonian  $H_{Sun}$  describing the interactions between Io and Callisto under the Sun's perturbative effects is obtained from [3]

$$H_{Sun} = -\frac{Gm_i m_\odot}{a_\odot} \left\{ \frac{1}{2} b_{1/2}^{(0)} \left( \frac{a_i}{a_\odot} \right) - 1 + \frac{1}{8} \frac{a_i}{a_\odot} b_{3/2}^{(1)} \left( \frac{a_i}{a_\odot} \right) (e_i^2 + e_\odot^2) \right\} \quad (6)$$

Here, the subscript  $i$  indicates that the orbital elements are associated with either Io or Callisto,  $e_i$  and  $e_\odot$  represent the orbital eccentricities of Io or Callisto and Jupiter's orbit around the Sun, respectively. The terms  $b_s^{(j)}(\alpha)$  where  $\alpha = a_i/a_{Sun}$ , denote Laplace coefficients [3], defined as

$$b_s^{(j)}(\alpha) = \frac{1}{\pi} \int_0^{2\pi} \frac{\cos(j\psi) d\psi}{(1 - 2\alpha \cos \psi + \alpha^2)^s} \quad (7)$$

Since  $\alpha \ll 1$ , the equation (7) is approximated using a binomial series expansion up to the second degree, yielding

$$b_{1/2}^{(0)}(\alpha) = 2 + \frac{1}{2} \alpha^2 \quad (8)$$

$$b_{3/2}^{(1)}(\alpha) = 3\alpha \quad (9)$$

Only terms up to the second degree in  $\alpha$  are considered, as higher-order terms are negligible.

To quantify the effect of solar perturbation, we consider the time derivative of the canonical momentum in Hamilton's equations,  $\dot{p} = -\partial H / \partial q$ , which corresponds to the

perturbative force acting on the celestial bodies. The only relevant force is gravitational, and solar perturbation remains within the Newtonian limit, hence differentiating equation (6) with respect to  $a_i$  yields the perturbative force exerted by the Sun on Io and Callisto:

$$F_{\text{perturb}}^{(\text{Sun})} = \frac{Gm_i m_s}{2a_{\odot}^2} \left[ \frac{a_i}{a_{\odot}^2} + \frac{3a_i(e_i + e_{\odot})}{8} \left( \frac{1}{a_{\odot}^2} + \frac{1}{a_{\odot}} \right) \right] \quad (10)$$

The total force,  $F_T$ , acting on Io or Callisto is then the sum of the gravitational attraction from Jupiter and the perturbative force from the Sun:

$$F_T = \frac{M_J m_i}{a_i^2} + F_{\text{perturb}}^{(\text{Sun})} \quad (11)$$

Setting equation (11) equal to the required centripetal force and solving for Jupiter's mass yields the modified mass estimate

$$\tilde{M}_J = \frac{4\pi^2 a_i^3}{GT^2} - \frac{m_{\odot} a_i^2}{2a_{\odot}^2} \left[ \frac{a_i}{a_{\odot}^2} + \frac{3a_i(e_i + e_{\odot})}{8} \left( \frac{1}{a_{\odot}^2} + \frac{1}{a_{\odot}} \right) \right] \quad (12)$$

By substituting the semi-major axes and masses of Io and Callisto and the parameters given in [2], the perturbative influence of the Sun on the calculation of Jupiter's mass can be evaluated. The results indicate that this perturbative effect is negligibly small.

### B. Mutual Satellites' Interactions

The Jovian system, or Jupiter's planetary system, consists of 95 satellites [8], all of which gravitationally interact with one another. Among them, the Galilean moons exert the most significant gravitational influence due to their larger masses. The perturbation effects caused by these moons can be evaluated using a similar approach to the one outlined in the previous section.

The Hamiltonian for the direct part (direct and effective gravitational interaction between the moons) of the disturbing function is given by:

$$H_{\text{int}} = -\frac{Gm_i m_{\text{Eu}}}{|a_{\text{Eu}} - a_i|} - \frac{Gm_i m_{\text{Gan}}}{|a_{\text{Gan}} - a_i|} - \frac{Gm_i m_j}{|a_i - a_j|} \quad (13)$$

where  $m_{\text{Eu}}$  and  $a_{\text{Eu}}$  are the mass and semi-major axis of Europa, and  $m_{\text{Gan}}$  and  $a_{\text{Gan}}$  correspond to that of Ganymede. The subscripts  $i$  and  $j$  represent the elements associated to Io or Callisto. The indirect term accounts for the gravitational influence of the moons on Jupiter. However, since Jupiter's mass is vastly greater than that of its moons, their effect on its motion is negligible, allowing this term to be ignored. Differentiating equation (13) with respect to  $a_i$  gives the perturbative force acting on Io or Callisto:

$$F_{\text{perturb}}^{(\text{satellite})} = \frac{Gm_i m_{\text{Eu}}}{|a_i - a_{\text{Eu}}|^3} (a_i - a_{\text{Eu}}) + \frac{Gm_i m_{\text{Gan}}}{|a_i - a_{\text{Gan}}|^3} (a_i - a_{\text{Gan}}) + \frac{Gm_i m_j}{|a_j - a_i|^3} (a_j - a_i) \quad (14)$$

The total force acting on Io or Callisto is thus the sum of Jupiter's gravitational attraction and the perturbative forces exerted by the other Galilean moons. This modifies the estimated mass of Jupiter, expressed as:

$$M'_J = \frac{4\pi^2 a_i^3}{GT^2} - \frac{a_i^2 m_{\text{Eu}}}{|a_i - a_{\text{Eu}}|^3} (a_i - a_{\text{Eu}}) - \frac{a_i^2 m_{\text{Gan}}}{|a_i - a_{\text{Gan}}|^3} (a_i - a_{\text{Gan}}) - \frac{a_i^2 m_j}{|a_j - a_i|^3} (a_j - a_i) \quad (15)$$

By substituting the relevant parameters into equation (15), the perturbative effect can be quantified. For Io, the perturbation introduces a 0.01% deviation in the mass estimate, whereas for Callisto, the effect is 0.06%. This suggests that Callisto's mass estimate is more affected by the gravitational perturbations of the other Galilean moons than Io's.

### C. Jupiter's Oblateness

Jupiter has the fastest rotation speed in the solar system, completing one rotation in just 9.93 hours [8]. This rapid spin causes the planet to bulge at the equator, resulting in a deviation from a perfect spheroidal shape. Consequently, Jupiter's mass distribution is not uniform, meaning it no longer strictly adheres to the assumptions of Newton's law of universal gravitation. This oblateness introduces perturbations [7] to its potential energy,  $\phi$ , which can be approximated using series expansion.

To simplify the calculation, Jupiter is assumed to have a constant density throughout the volume and exhibit axial symmetry. The simplified potential energy, evaluated at the equatorial plane [14], is expressed as

$$\tilde{\phi}(a_i) = -\frac{GM_J m_i}{a_i} \left[ 1 - \sum_{n=1}^{\infty} (-1)^n \frac{(2n)!}{(2^n n!)^2} J_{2n} \left( \frac{R}{a_i} \right)^{2n} \right] \quad (16)$$

where  $J_{2n}$  is the dimensionless multipole moment coefficient that quantifies deviations in Jupiter's gravitational field due to its oblateness, and  $R$  is the mean equatorial radius. The subscript  $i$  retains its usual meaning, as defined in the discussion previously. Since  $J_2$  is the dominant term [14], the series in equation (16) is truncated at the first degree, yielding

$$\tilde{\phi}(a_i) = -\frac{GM_1 M_2}{a_i} \left[ 1 + J_2 \left( \frac{R}{a_i} \right)^2 \right] \quad (17)$$

By taking the derivative of equation (17), the gravitational force acting on Io and Callisto is expressed as

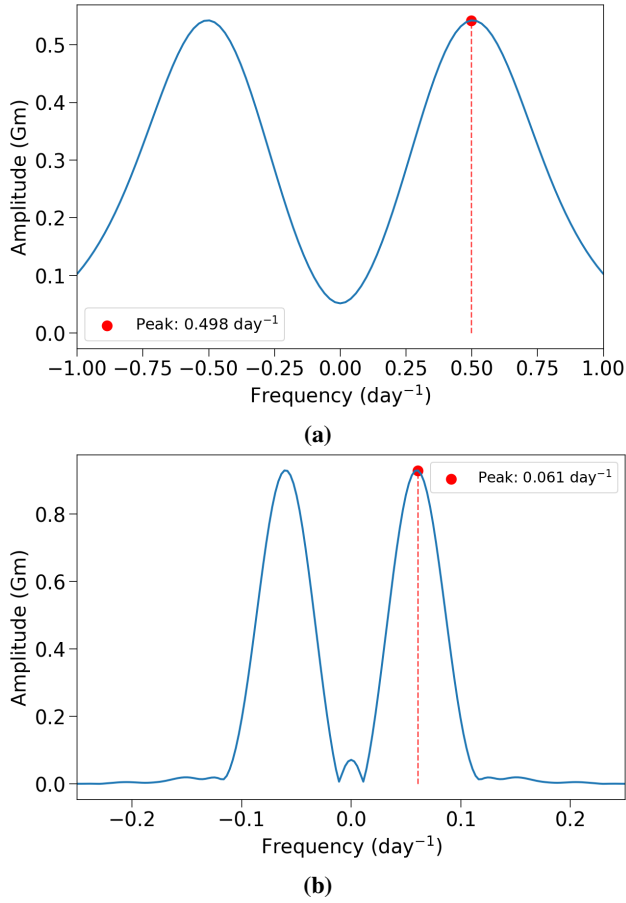
$$F(r) = \frac{d\tilde{\phi}}{da_i} = \frac{GM_J m_i}{a_i^2} \left( 1 + \frac{3J_2 R^2}{a_i^2} \right) \quad (18)$$

Equating equation (18) with the centripetal force, the modified mass of Jupiter,  $\tilde{M}_J$ , is

$$\tilde{M}_J = \left( \frac{a_i^2}{a_i^2 + 3J_2 R^2} \right) M_J \quad (19)$$

Using  $J_2 = 0.014736$  for Jupiter [8], the deviations in its estimated mass due to oblateness are found to be approximately 0.12% for Io and 0.0063% for Callisto, based on their respective semi-major axes. However, this value represents a lower bound for the perturbation, as Jupiter has a density gradient and is not perfectly axially symmetric. Factors such as its asymmetric magnetosphere [5] and atmospheric disturbances — most notably the Great Red Spot, which is active in the southern hemisphere — further contribute to deviations from the simplified model.

## V. DISCUSSION



**FIG. 3:** (a) Fourier transform of Io's orbital motion graph, showing a Gaussian-like structure. (b) Fourier transform of Callisto's orbital motion graph, showing a relatively narrower distribution compared to Io's.

The perturbation caused by other planets is neglected in this paper, as it primarily induces long-term effects [7] on the orbits of Io and Callisto rather than the short-term effects. Consequently, the dominant perturbative effects considered are solar perturbation, mutual interactions between the satellites, and Jupiter's oblateness. report

Solar perturbation affects Io and Callisto similarly, despite the Sun being the most massive body in the Solar System. While mutual satellite interactions exert a stronger perturbative effect on Io than on Callisto, this difference is significantly smaller than the effect of Jupiter's oblateness. This indicates that Jupiter's oblateness is the dominant perturbation influencing both moons' orbits. Given their proximity to Jupiter and its immense mass, this result

is expected. Io, being the closest Galilean moon to Jupiter, experiences the strongest perturbation due to its oblateness. Moreover, this difference suggests that the potential sine-wave pattern observed in the residuals of Io's displacement-time graph is at least partially attributed to Jupiter's oblateness, though further data points are needed for confirmation.

The Fourier transform of a pure sine wave in a finite frequency domain is expected to yield a Lorentzian function. FIG. 3 shows the Fourier transforms of the moons' orbital motion. Io's Fourier transform in FIG. 3(a) exhibits a Gaussian-like structure and is broader than that of Callisto's in FIG. 3(b). In contrast, Callisto's Fourier transform more closely follows a Lorentzian function, despite not being perfectly smooth at the origin and tails. This discrepancy suggests the presence of a missing term in the model, leading to a broadening effect in Io's Fourier spectrum. A revision of the model is necessary to achieve a more accurate representation.

## VI. CONCLUSION

The calculations show that using the masses and semi-major axes of Io and Callisto yields different results:  $(2.0 \pm 0.5) \times 10^{27}$  kg for Io and  $(1.91 \pm 0.03) \times 10^{27}$  kg for Callisto. Compared to NASA's reported value of  $1.8981(8) \times 10^{27}$  kg [11], Callisto provides a more accurate estimate, as it experiences less perturbation from Jupiter's oblateness than Io. Therefore, the hypothesis is accepted.

## VII. REFERENCES

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## VIII. ERROR APPENDIX

The equations used in this section are based on Measurement and their Uncertainties by Hughes and Hase [4].

### A. Error Bars

The uncertainties in the distances between Jupiter and its moons (y error bars) arise from determining their centres. FIG. 1 was cropped to isolate Jupiter, Io, and Callisto for error analysis. To remove unnecessary colours, the colour map was set to greyscale, and the colour scale was reverted to its default setting, rendering the images in grey. The images were then imported into a Python script to extract the pixel intensities in both horizontal and vertical directions, and 3D Gaussian plots were generated to calculate the standard deviations along the x-axis,  $\sigma_x$  and y-axis,  $\sigma_y$ .

A discrepancy was found between the actual size of the image and its size when opened by an image viewer. When the images were sent to the Python script, the number of pixels read by the script was scaled up, leading to deviated standard deviations. To correct this, another Python script was run to determine the number of pixels in the scaled-up images, and the real number of pixels was manually counted. The scaling factor was calculated by dividing the number of pixels in the scaled-up images by the real number of pixels.

The image of Jupiter was over-exposed, creating a top on the 3D Gaussian, as shown in FIG. 4(a). Hence, the 3D Gaussian was extrapolated from the tail to the peak to obtain accurate standard deviations, as shown in FIG. 4(b).

The final standard deviations,  $\sigma_f$  in pixels were calculated as the quadrature of  $\sigma_x$  and  $\sigma_y$

$$\sigma_f = \sqrt{\sigma_x^2 + \sigma_y^2} \quad (20)$$

The final standard deviations are listed below:

| Jupiter/moons | $\sigma_x$ (pixels) | $\sigma_y$ (pixels) | $\sigma_f$ (pixels) |
|---------------|---------------------|---------------------|---------------------|
| Jupiter       | 12.7                | 9.42                | 15.8                |
| Io            | 2.25                | 1.89                | 2.94                |
| Callisto      | 2.32                | 2.05                | 3.10                |

**TABLE I:** Table of  $\sigma_x$ ,  $\sigma_y$  and  $\sigma_f$  of Jupiter and its moons

Similar to the conversion and equation (4) suggested in Methodology section, the y-error bars are calculated as

$$\alpha = \frac{0.983(\sigma_f^{(J)} + \sigma_f^{(i)})r}{206265} \quad (21)$$

where  $r$  and superscript  $i$  have the usual meanings.

### B. Uncertainties in Orbital Periods and Semi-major Axes

The uncertainties in the best-fit parameters are taken to be the standard errors,  $\alpha$ , which can be obtained by taking the square root of the diagonal elements of the covariance matrix,  $C$

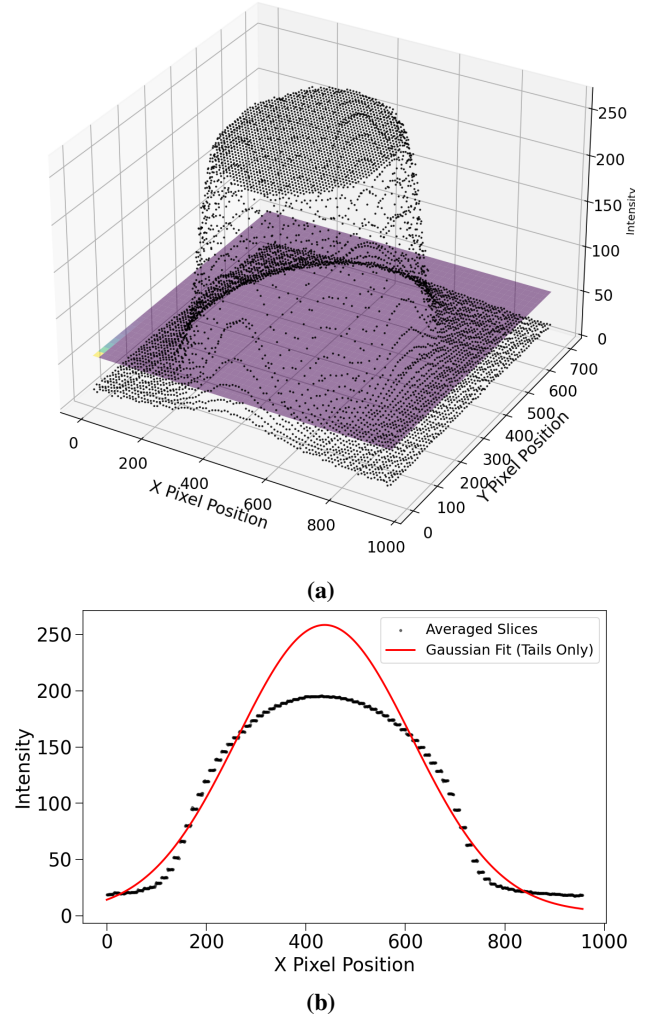
$$\alpha_j = \sqrt{C_{jj}} \quad (22)$$

where  $\alpha_j$  is uncertainty of the j-th parameter, and  $C$  is the inverse of curvature matrix,  $A$ , defined by

$$A_{jk} = \frac{1}{2} \frac{\partial^2 \chi^2}{\partial a_j \partial a_k} \quad (23)$$

where  $a_j$  and  $a_k$  are the j-th and k-th parameter, respectively.

Standard errors of parameters  $A$  and  $B$  in equation (5) correspond to the uncertainties in the semi-major axes of the moons and the orbital speed of their orbits, respectively. The uncertainties in the orbital period and mass of Jupiter can be obtained through standard error propagation.



**FIG. 4:** (a) The CCD of the telescope was saturated when taking images of Jupiter, leading to a constant intensity, which resulted in an incomplete 3D Gaussian model. (b) Multiple slices of the 3D Gaussian are taken and average to remove noise. The Y pixel position was removed.

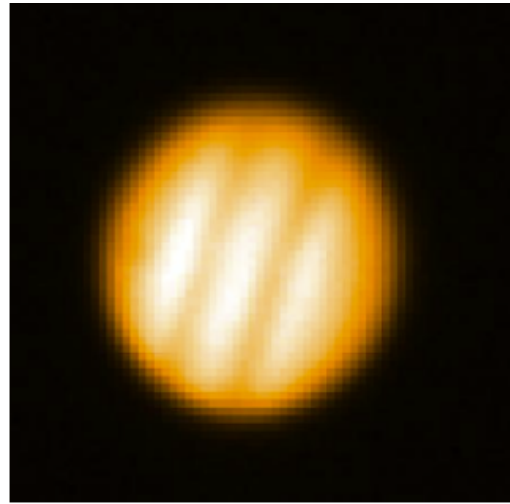
## IX. LAY SUMMARY FOR GENERAL AUDIENCE

Jupiter is the most largest planet in our solar system. It has a complex planetary system of 95 moons, with four of them—Io, Europa, Ganymede, and Callisto—being the largest. These four moons are known as the Galilean moons.

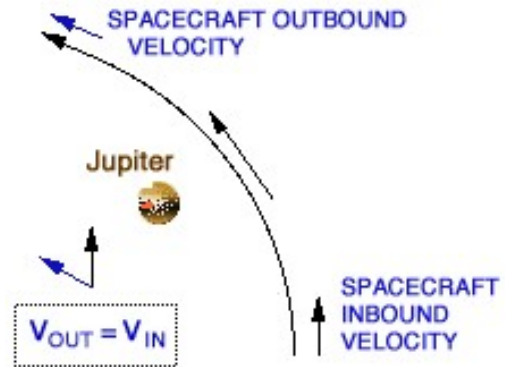
One way to estimate Jupiter's mass is by studying the Galilean moons, using equation (3). However, this equation doesn't take into account external factors, such as the Sun's gravity, the mutual effects between the moons, and oblateness of Jupiter (shown in FIG.5). This paper explores how these factors, or perturbations, affect the accuracy of estimating Jupiter's mass by tracking the positions of Io, the closest moon, and Callisto, the farthest.

Telescope images of Io and Callisto were analysed to obtain their distances from Jupiter,  $a$  and time taken for them to complete an orbit,  $T$ . The data is then substituted in equation (3) to find the mass estimate of Jupiter. Callisto provided a more accurate result than Io, as it is less affected by the overall perturbation.

Understanding Jupiter's mass and how its moons are influenced by these factors is crucial for space exploration. Accurate calculations are needed for planning spacecraft and probe missions to avoid collision with Jupiter, especially for gravity assist manoeuvres (shown in FIG.6).



**FIG. 5:** Due to Jupiter's rapid rotation, its equator appears bulged, making it an oblate spheroid rather than a perfect sphere.



**FIG. 6:** The spacecraft approaches Jupiter and is redirected at a higher speed. It is a manoeuvre that helps accelerate and conserve fuel. Credit: The Worlds of David Darling

## **X. GENERATIVE AI STATEMENT**

In the production of this scientific paper, ChatGPT was used to correct grammar, improve the coherence of sentences, fix spelling mistakes, and enhance the flow of passages. The work and content are entirely my own, and the reference sources were selected without reliance on ChatGPT.